RODRIGO GOMES DE ARAÚJO

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Education

Master's **Sep 2025 – Present (Jul 2027)**
Computer Science *University of Porto (FCUP)*

Bachelor's **Sep 2022 – Jul 2025**
Computer Science and Engineering *University of Porto (FEUP)*

Research

CaST (Cryptographically-aware Session Types) **Jul 2026 – Present (Jul 2027)**
Master's Thesis · Advisors: Pedro Ângelo, Bernardo Portela *FreeST, Haskell, Type Systems, Applied Cryptography*

- Extending session types to capture **cryptographic resource usage**, so that misuse in protocols is rejected at compile-time.
- Formalizing the extension with a **soundness proof**, integrating it into the FreeST compiler and exploring applications in **smart contracts** and protocol specification.
- Built a **Kademlia DHT in FreeST** and the synthax highlighting extension for Zed.

Experience

Murph (Devtools Startup) **Jul 2026 – Present**
Co-founder & Lead Engineer *Rust, Gitoxide, Ed25519, IPLD/DAG-CBOR, Crlu, Serde*

- Co-founded and lead a **3-person startup**. **Primary author of the codebase** and owner of the architecture. Murph merges **version control, project management and CI** into one system.
- Designed the **security and trust model**. **Ed25519-signed approvals** bound to the exact action, **tiered authority** for agents and humans and a **tamper-evident history** linking every change in production to its spec, verification and approval.
- Fully built in **Rust**, with the change lifecycle enforced by **typestate** and **deterministic, concurrent verification** of immutable versions over a Git underlay.

Coursework

PoW Blockchain for Decentralized Auctions
Data Systems Security · 19/20 *Rust, Tokio, LibP2P, Docker, Blake2b, Ed25519*

- **Research on the state of the art** of Bitcoin, Ethereum and Solana, Kademlia and churn/attack literature before building.
- **Kademlia DHT** overlay for decentralized network construction, **GossipSub** block propagation, **chain synchronization and fork resolution**.
- PoW chain with **Blake2b hashing**, **Ed25519 signatures** and **Merkle Trees** for transaction validation, used as a public ledger for an **auction client**.
- Docker **adversarial network simulation** with bootstrap, well behaved and **malicious nodes**.

Distributed Protocols
Distributed Systems · 19/20 *Rust, Tokio, gRPC (Tonic), Docker*

- **Totally ordered multicast** with **Lamport clocks**, defended against timestamp reuse, backdating and far-future attacks.
- **Token ring mutual exclusion** with **failure detection** and automatic ring healing.
- **Push-pull gossip** with exponential **anti-entropy** for state convergence.

Portuguese Elections Data Platform
Advanced Topics in Databases · 18/20 *PostgreSQL, PostGIS, Python, OCaml*

- **ETL pipeline** normalizing **10 years** of official results (8 elections) from **inconsistent** government spreadsheets: layout detection, **entity resolution** of parties, **idempotent** loads with **SHA-256 provenance**.
- **Automated reconciliation** against official totals (votes, turnout, seats) and **invariant checks** that must return zero. Triggers reject bad writes and **D'Hondt** seats are recomputed in SQL as a cross-check.
- **Star-schema warehouse** with **CUBE/ROLLUP**, **window functions** and **recursive drill-down** to 3,000+ parishes. Vote-swing analytics and **in-database PostGIS** maps served by an **OCaml** API to a **Bonsai** UI.

Awards

2nd Place in IEEE RetroJam 2025 (48h game from scratch, Rust) | IEEE UP Student Branch **Oct 2025**

1st Place National Robotics Championship · RoboCup 2016 (Leipzig) | Instituto Politécnico de Bragança **May 2016**

Technical Skills

Languages: Rust, Python, C/C++, Java, SQL, Haskell, OCaml, TypeScript
Distributed Systems & Cryptography: Tokio, libp2p (Kademlia, GossipSub), gRPC, Ed25519, Blake2b, Merkle Trees
Data & Infrastructure: PostgreSQL, PostGIS, pandas, NumPy, scikit-learn, FastAPI, Docker, Terraform, Git, UNIX/Linux
Web: React, Dioxus, Bonsai, Dream, HTML5, CSS
Spoken Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate)